

Appointment of PKL – Duration for 3 months (May – July 2026)

Task 1: ANFIS and ANN Modelling for EDM Datasets

PKL is required to conduct simulation and modelling using Adaptive Neuro-Fuzzy Inference System (ANFIS) and Artificial Neural Network (ANN) techniques for the provided EDM Dataset 1 and EDM Dataset 2.

Detailed Responsibilities:

1. Data Preparation

- i. Review and understand the structure of Dataset 1 and Dataset 2

2. ANFIS Modelling

- i. Develop ANFIS models for both datasets.
- ii. Select suitable membership functions and training parameters.
- iii. Run training and testing processes.
- iv. Record performance metrics such as RMSE, MAE, R^2 , or other relevant indicators.

3. ANN Modelling

- i. Develop ANN models for both datasets.
- ii. Determine suitable network architecture (hidden layers, neurons, activation functions).
- iii. Conduct training, validation, and testing.
- iv. Compare prediction accuracy and model performance.

4. Results Compilation

- i. Prepare and organize all simulation outputs in Microsoft Excel.
- ii. Include tables for predicted values, actual values, error values, and model performance metrics.
- iii. Ensure all files are clearly labelled and systematically arranged.

5. Graphical Outputs

- i. Generate graphical representations of the results (comparison between ANFIS and ANN), such as:
 - Predicted vs Actual values
 - Error convergence plots
 - Training performance graphs
- ii. Insert all graphs into the Excel file and separate report folder.

Expected Deliverables (all output should be submitted in the following link:

<https://drive.google.com/drive/folders/10ELXq4YrIFhilQV8-Fj7Jtp02ZKgJJFj?usp=sharing>)

1. Completed ANFIS simulation results for Dataset 1 & Dataset 2
2. Completed ANN simulation results for Dataset 1 & Dataset 2
3. Excel file containing all results and analysis
4. Graphical outputs of simulations

Prediction papers:

https://drive.google.com/file/d/1d8DW2rlml4ynSoQa8xlz3BenqE5V3CEM/view?usp=drive_link

https://drive.google.com/file/d/1ErBAqZJQ7GOzMVkHEwWUxXkx-hjS3Pkv/view?usp=drive_link

https://drive.google.com/file/d/13c6EX6k6KTurNw8B9uGqHfHBknLpV5Xt/view?usp=drive_link

Prepared by,

Dr. Nurezayana Zainal